

FICO Score Rating Mapping Using Quantization

Introduction

Banks and financial institutions use FICO scores to measure the creditworthiness of borrowers. A higher FICO score indicates that the borrower is less likely to default on a loan, while a lower FICO score indicates higher risk.

Instead of working with hundreds of individual FICO scores, banks often group similar scores into categories called ratings. This process is known as quantization or bucketing.

The objective of this project is to convert FICO scores into credit ratings by creating score ranges (buckets). A lower rating number represents a better credit profile and lower risk of default.

Objective

The main objectives of this project are:

- Group FICO scores into rating categories.
 - Create a simple rating system for borrowers.
 - Analyze default behavior within each rating group.
 - Build a rating map that can be used for future borrowers.
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Problem Statement

The dataset contains information about borrowers, including their FICO scores and loan default history.

Using this information, we need to:

1. Divide FICO scores into buckets.
 2. Assign a rating to each bucket.
 3. Ensure borrowers with similar credit risk are grouped together.
 4. Create a rating system that can be used for future datasets.
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What is Quantization?

Quantization means converting a large number of values into a smaller number of groups.

For example:

Instead of using:

- 620
- 621
- 622
- 623

We can group them into:

- Rating 4

This makes analysis easier and helps banks classify borrowers more effectively.

Method Used

In this project, FICO scores were divided into multiple buckets using quantile-based grouping.

This approach automatically creates groups containing a similar number of borrowers.

Each bucket was assigned a rating:

- Rating 1 = Lowest Risk
 - Rating 2 = Low Risk
 - Rating 3 = Medium Risk
 - Rating 4 = High Risk
 - Rating 5 = Highest Risk
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Rating Structure Example

FICO Score Range	Rating
760 - 850	1
710 - 759	2
660 - 709	3
610 - 659	4
Below 610	5

Note: Actual ranges may vary depending on the dataset.

Data Analysis

The FICO score distribution was analyzed to understand how borrowers are spread across different score ranges.

After creating the buckets, the default rate for each rating group was calculated.

This helped determine whether lower FICO scores are associated with higher default rates.

Rating Mapping Function

A rating function was created that takes a borrower's FICO score as input and returns the corresponding credit rating.

Example:

FICO Score = 780

Output:

Rating = 1

This allows future borrowers to be assigned a rating automatically.

Results

The rating system successfully grouped borrowers into different risk categories.

The analysis showed that:

- Higher FICO scores generally had lower default rates.
 - Lower FICO scores generally had higher default rates.
 - The rating system provided a simple way to classify borrower risk.
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Business Benefits

This rating model can help banks:

- Evaluate borrower risk quickly.

- Improve loan approval decisions.
 - Reduce credit losses.
 - Standardize credit assessment.
 - Monitor high-risk borrowers.
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Conclusion

This project successfully converted FICO scores into credit ratings using a quantization approach. By grouping borrowers into rating categories, the model simplifies credit risk assessment and makes borrower evaluation easier.

The results demonstrate that borrowers with higher FICO scores generally have lower probabilities of default, while borrowers with lower FICO scores carry higher credit risk. This rating system can be used as a foundation for future credit risk models and lending decisions.